Chapter 6
Data Warehouse and Data Mining

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Outline

- Data Warehousing
  - ✓ Introduction
  - ✓ Benefits
  - ✓ Online Transaction Processing (OLTP) and Data Warehousing
- Data Mining
  - ✓ Introduction
  - ✓ Data Mining Techniques

What is Data Warehouse?

- Data warehouse refers to a **data repository** that is maintained separately from an organization's operational databases.
- It is known as **decision support database.**
- A data warehouse is a **subject-oriented, integrated, time varying, non-volatile** collection of data that is used primarily in organizational decision-making process.
- It contains a collection of **technologies** (hardware, database software, and tools), that enabling the *knowledge worker* (executive, manager, analyst) to make better and faster decisions.
- Support information processing by providing a solid platform of consolidated, historical data for analysis.

Major Data Warehouse features - Subject Oriented

- Data Warehouse **is organized** around major subjects such as customer, supplier, product, and sales.
- Data Warehouse **focuses** on the modeling and analysis of data for decision makers **instead of** concentrating on the day-to-day operations and transaction processing of an organization.
- Data Warehouse **provides** a simple and concise view around particular subject issues by excluding data that is not useful in the **decision support** process.

Major Data Warehouse features – Integrated

- Data Warehouse is constructed by **integrating** multiple heterogeneous sources:
  - ✓Relational databases
  - ✓Flat files
  - ✓Online transaction records
- Data Warehouse **applies** data cleaning and data integration techniques to ensure:
  - ✓consistency in naming conventions
  - ✓encoding structures
  - ✓attribute measures, etc.…

Major Data Warehouse features – Time-Variant

- The time horizon for a data warehouse is significantly longer than that of operational systems.
  - – Operational database: current value data
  - – Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years).
- Every key structure in the data warehouse contains, either implicitly or explicitly, a time element.
- But the key structure of operational data may or may not contain “time element”.

Major Data Warehouse features – Non-Volatile

- Physically separate **store of data** transformed from the **application data** found in the **operational** environment.
- Operational update of data does not occur in the data warehouse environment.
  - **Does not** require transaction processing, recovery and concurrency control mechanisms.
  - **Requires** only two **operations** in **data accessing**
    - ✓ Initial loading of data and **access** of data

“How are organizations using the information from data warehouses?”

- Many organizations use this information to support business decision-making activities, including

1. **Increasing customer focus**, which includes the analysis of customer buying patterns (such as buying preference, buying time, budget cycles).
2. **Repositioning products and managing product portfolios** by comparing the performance of sales by quarter, by year, and by geographic regions in order to fine-tune production strategies;
3. **Analyzing operations and looking for sources of profit**; and
4. **Managing customer relationships**, making environmental corrections, and managing the cost of corporate assets.

Benefits Data Warehouse

1. **Centralized Data Storage:** It consolidates data from multiple sources into a single repository, making it easier for organizations to manage and retrieve data.
2. **Historical Analysis:** They store historical data, facilitates analysis of historical trends, identifying patterns and predicting future outcomes.
3. **Business Intelligence Support:** Data warehouses serve as the backbone for BI tools, enabling advanced analytics, reporting, and visualization.
   - ✓ **Data-Driven Insights:** Powers BI tools, generating reports, dashboards, and visualizations for a deeper understanding of key performance indicators (KPIs).
4. **Better Decision-Making:** By providing a unified view of data, organizations can derive insights that support strategic decision-making.

Benefits Data Warehouse(cont'd)

5. **Improved Data Quality and Consistency:** Data warehousing processes often include cleaning and transforming data, which enhances its quality and consistency across the organization.
6. **Enhanced Query Performance:** Data warehouses are optimized for read-heavy operations, allowing for faster query performance and efficient data retrieval.
   - ✓ That means data warehouse is optimized for complex queries.
7. **Enhanced Insights:** Supports deeper dives into data, uncovering hidden relationships and gaining a comprehensive understanding of business operations.
8. **Scalability and Flexibility:** Designed to accommodate increasing/growing volume of data as businesses grow and data sources expand.

Operational Data Base Systems Vs Data Warehouses

❖**Operational database systems**

- The major task of **operational database systems** is **on-line transaction processing (OLTP)**.
- Handles day-to-day business transaction/operations, such as purchasing, inventory, banking, payroll, registration, and accounting.

❖ **Data warehouse systems**,

- Major task of data warehouse system is **online analytical processing(OLAP).**
- Serve users or knowledge workers(executive, manger, analyst) in the role of data analysis and decision making.

Operational Data Base Systems Vs Datawarehouse

- Operational Database (are finely tuned to support known OLTP workloads): access methods, indexing, Concurrency control, recovery mechanisms (e.g., locking and logging) are required to ensure the consistency and robustness of transactions.
  - Trying to execute complex OLAP queries against the operational databases would result in unacceptable performance
  - Relational databases put data into tables, while OLAP uses a multidimensional array representation(data cube).
- Data Warehouse (tuned for / to support OLAP): complex OLAP queries, multidimensional view, and consolidation.
- An OLAP query often needs read-only access of data records for summarization and aggregation. Data warehouses, are targeted for decision support.

Distinct features (OLTP vs. OLAP)

- **Users and system orientation**:
  - An OLTP system is ***customer-oriented*** and is used for transaction and query processing by clerks, clients, and information technology professionals.
  - An OLAP system is ***market-oriented*** and is used for data analysis by knowledge workers, including managers, executives, and analysts.
- **Data contents**:
  - An OLTP system manages **current data, detailed.**
  - An OLAP system manages large amounts of **historical data**, provides facilities for summarization and aggregation, and stores and manages information at different levels of granularity.
- **View**: current, local vs. evolutionary, integrated
- **Granularity** mean the lowest level of **information** to be **stored** in the fact table.

OLTP vs. OLAP

- **Database design**:
  - An OLTP system usually adopts an ***entity-relationship (ER)*** data model and an ***application-oriented*** database design.
  - An OLAP system typically adopts either a ***star*** or ***snowflake*** model and a ***subject oriented*** database design.
- **Access patterns**:
  - The access patterns of an OLTP system consist mainly of ***short, atomic transactions***.
    - Such a system requires concurrency control and recovery mechanisms.
  - However, accesses to OLAP systems are mostly ***read-only operations,*** although many could be complex queries

Table 4.1 Comparison of OLTP and OLAP Systems

Feature	*OLTP*	*OLAP*
Characteristic	operational processing	informational processing
Orientation	transaction	analysis
User	clerk, DBA, database professional	knowledge worker (e.g., manager, executive, analyst)
Function	day-to-day operations	long-term informational requirements decision support
DB design	ER-based, application-oriented	star/snowflake, subject-oriented
Data	current, guaranteed up-to-date	historic, accuracy maintained over time
Summarization	primitive, highly detailed	summarized, consolidated
View	detailed, flat relational	summarized, multidimensional
Unit of work	short, simple transaction	complex query
Access	read/write	mostly read
Focus	data in	information out
Operations	index/hash on primary key	lots of scans
Number of records accessed	tens	millions
Number of users	thousands	hundreds
DB size	GB to high-order GB	$\geq$ TB
Priority	high performance, high availability	high flexibility, end-user autonomy
Metric	transaction throughput	query throughput, response time

Why Data Mining



We are data rich, but information poor

- Huge volumes of data are accumulated in databases and data warehouses.
- Huge volumes of data also come from WWW and data streams
  - (video surveillance, telecommunication, and sensor networks)
- Effective and efficient analysis of data in different forms becomes a challenging task.
- Fast-growing, tremendous amount of data, collected and stored in large and numerous data repositories, has far exceeded our human ability for comprehension without powerful tools
- Decision-making is done according to **information** (not data)

“We are Drowning in Data…”



Predict

- Interests and behavior of mankind



Facebook

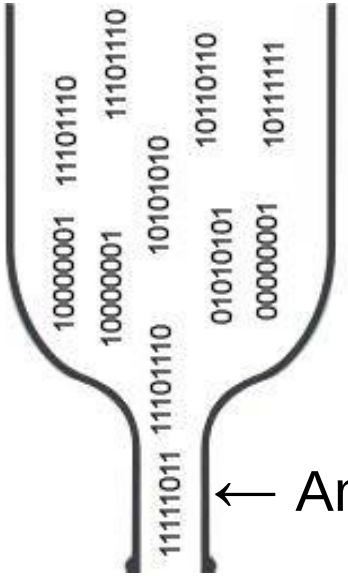
- 4 Petabyte of new data generated every day
- over 300 Petabyte in Facebook‘s data warehouse

Predict

- Interests and behavior of over one billion people

Amount of Data Created Daily (2024) - Approximately **402.74 petabytes** of data are created each day

“…but starving for knowledge!”



← Amount of data that is collected

← Amount of data that can be looked at by humans

We are interested in the patterns, not the data itself! Data Mining methods help us to

- discover interesting patterns in large quantities of data
- take decisions based on the patterns

What is Data Mining?

❖**Data mining** is the process of discovering interesting **patterns** and **knowledge** from large amounts of data.

❖Patterns must be:

- ✓ valid, novel, potentially useful, understandable

❖Popularly known as ***Knowledge Discovery*** *in Data* (KDD).

❖ More appropriately named “knowledge mining from data”

❖Alternative names

- Knowledge discovery in data (KDD), knowledge extraction, pattern analysis, data archeology, data dredging, etc.

Knowledge Discovery in Data: Example

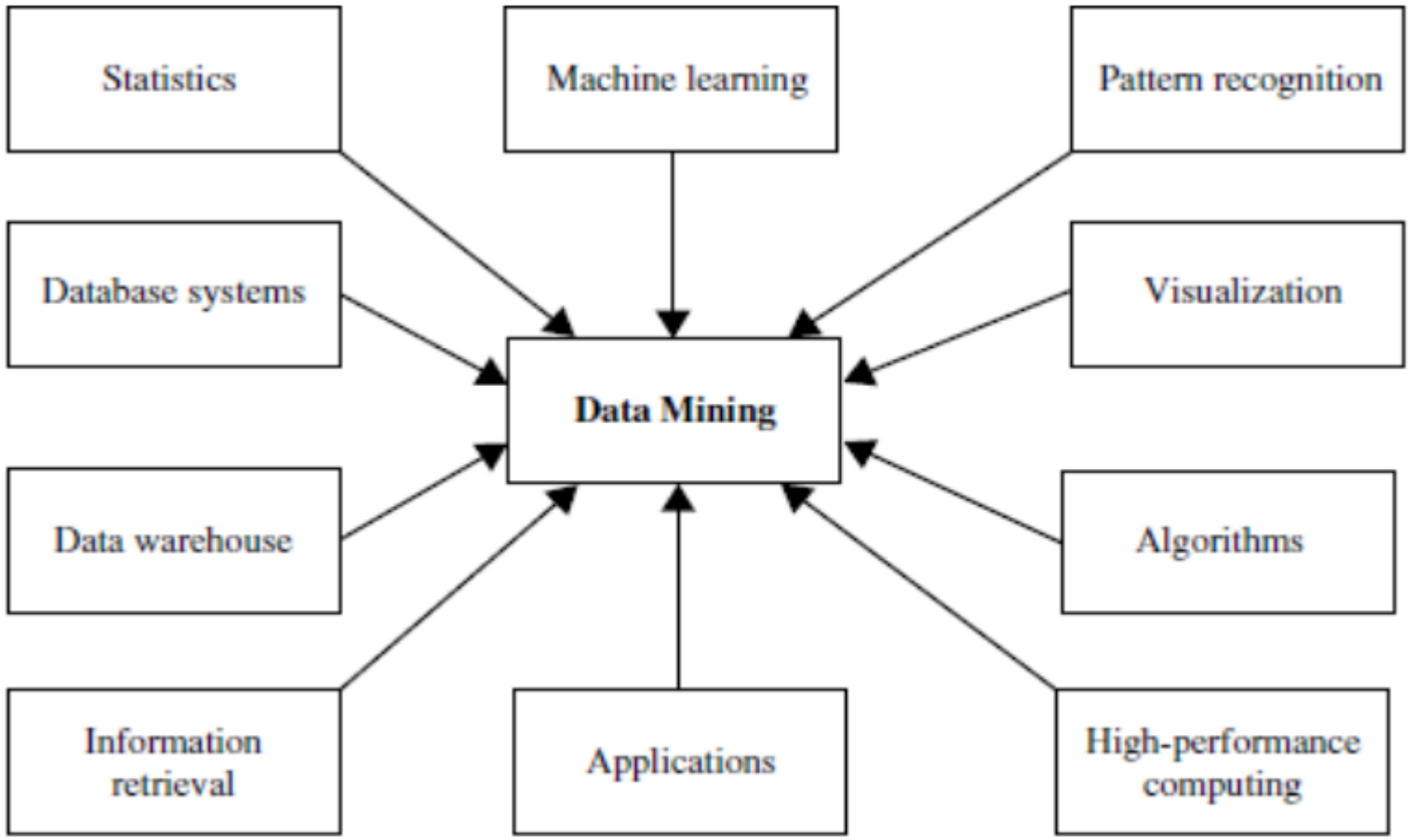
Example 1: Analysis of purchases in a supermarket

customer1	pizza	beer	cheese	bread	chips
customer2	milk	bread	ham	cigaretts	
customer3	yoghurt	sugar	flour	cornflakes	napkins
customer4	shampoo	beer	chips	newspaper	pizza
xustomer5	chips	coffee	beer	pizza	cream
customer6	jam	rolls	butter	beer	
...					



If pizza and beer are in one purchase, it is likely that chips are also in that purchase.

Which Technologies Are Used?



Data mining adopts techniques from many domains.

Statistics studies the collection, analysis, interpretation or explanation, and presentation of data. Data mining has an inherent connection with statistics.

Machine learning investigates how computers can learn (or improve their performance) based on data.

Data Mining and Knowledge Discovery Process

- A view from typical database systems and data warehousing communities
- Data mining plays an essential role in the knowledge discovery process

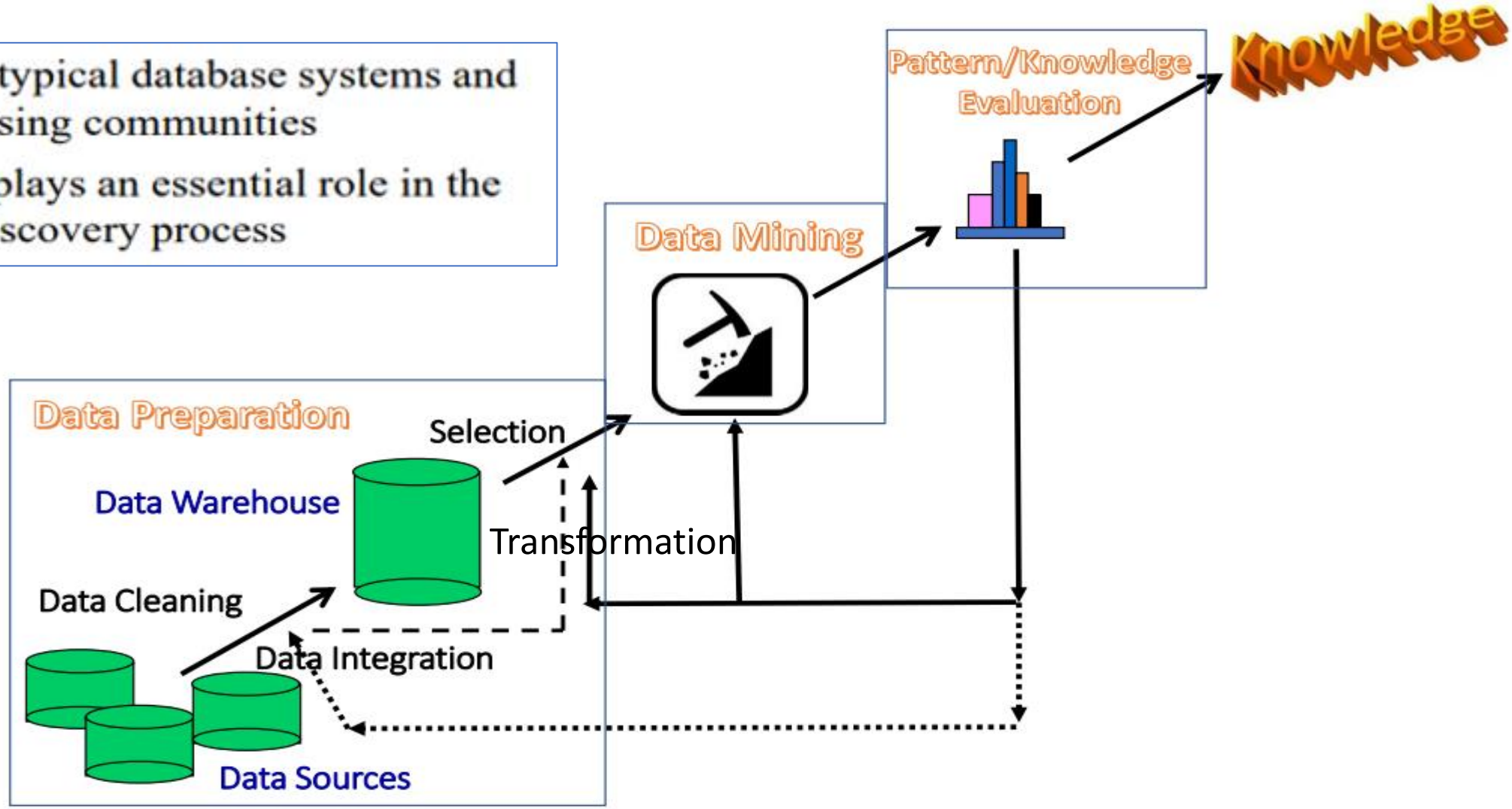


Figure: Data mining: An essential step in the process of knowledge discovery.

Steps involved in Knowledge Discovery Process

1. Data Preparation

a. **Data cleaning**: it is a phase in which **noise, inconsistent** and **irrelevant data** are removed from the collection.

b. **Data integration**: at this stage, multiple data sources, often heterogeneous, may be combined in a common source.

c. **Data selection**: at this step, the **data relevant** to the **analysis task** is retrieved from the **data collection(database)**. What data is potentially useful for the task at hand?

d. **Data transformation**: it is a phase in which the selected data is transformed and consolidated into forms appropriate for the mining procedure.

Steps involved in Knowledge Discovery Process

2. Data mining: crucial step where intelligent methods are applied in order to extract data patterns.

3. Pattern evaluation: In this step, data patterns are evaluated/interpreted.

4. Knowledge presentation: is the final phase in which the discovered/mined knowledge is visually represented to the user. This essential step uses visualization techniques to help users understand and interpret the data mining results.

What Kinds of Data Can Be Mined?

- Mining can be performed in a variety of data
- The most common forms of data for mining applications are database data, data warehouse data, and transactional data.
- Data mining can also be applied to other forms of data (e.g., data streams, text data, time-series(temporal data), multimedia data, WWW, graphs & networks and other data repositories).

What Kinds of Patterns Can Be Mined? Or Datamining Functionalities Or Datamining Task Primitives

– Data mining functionalities are used to specify the kinds of patterns to be found in data mining tasks. In general, such tasks can be classified into two categories:

1. Descriptive Tasks:

– Find human-interpretable patterns that describe the data.

- Goal: Find patterns in the data.
- Example: W*hich products are often bought together?*

2. Predictive Tasks:

– Use some variables to predict unknown or future values of other variables.

– Goal: Predict unknown values of a variable given observations (e.g., from the past)

– Example: W*ill a person click a online advertisement?* given her browsing history

– Machine Learning Terminology

- descriptive = unsupervised
- predictive = supervised

Data Mining Techniques

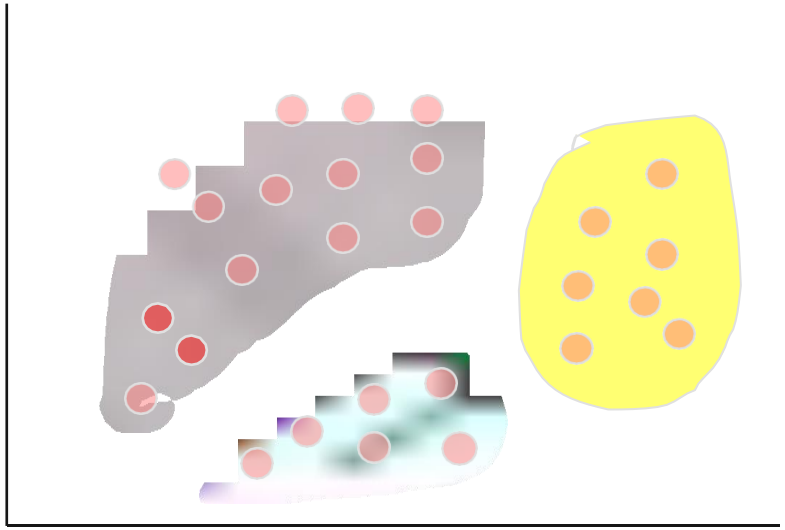
Data mining includes the utilization of refined data analysis tools to find previously unknown, valid patterns and relationships in huge data sets.

Major data mining techniques used are:

1. Clustering [Descriptive]
2. Classification [Predictive]
3. Regression [Predictive]
4. Association Rules [Descriptive]
5. Frequent Patterns [Descriptive]
6. Outlier analysis/– Deviation Detection [Predictive]

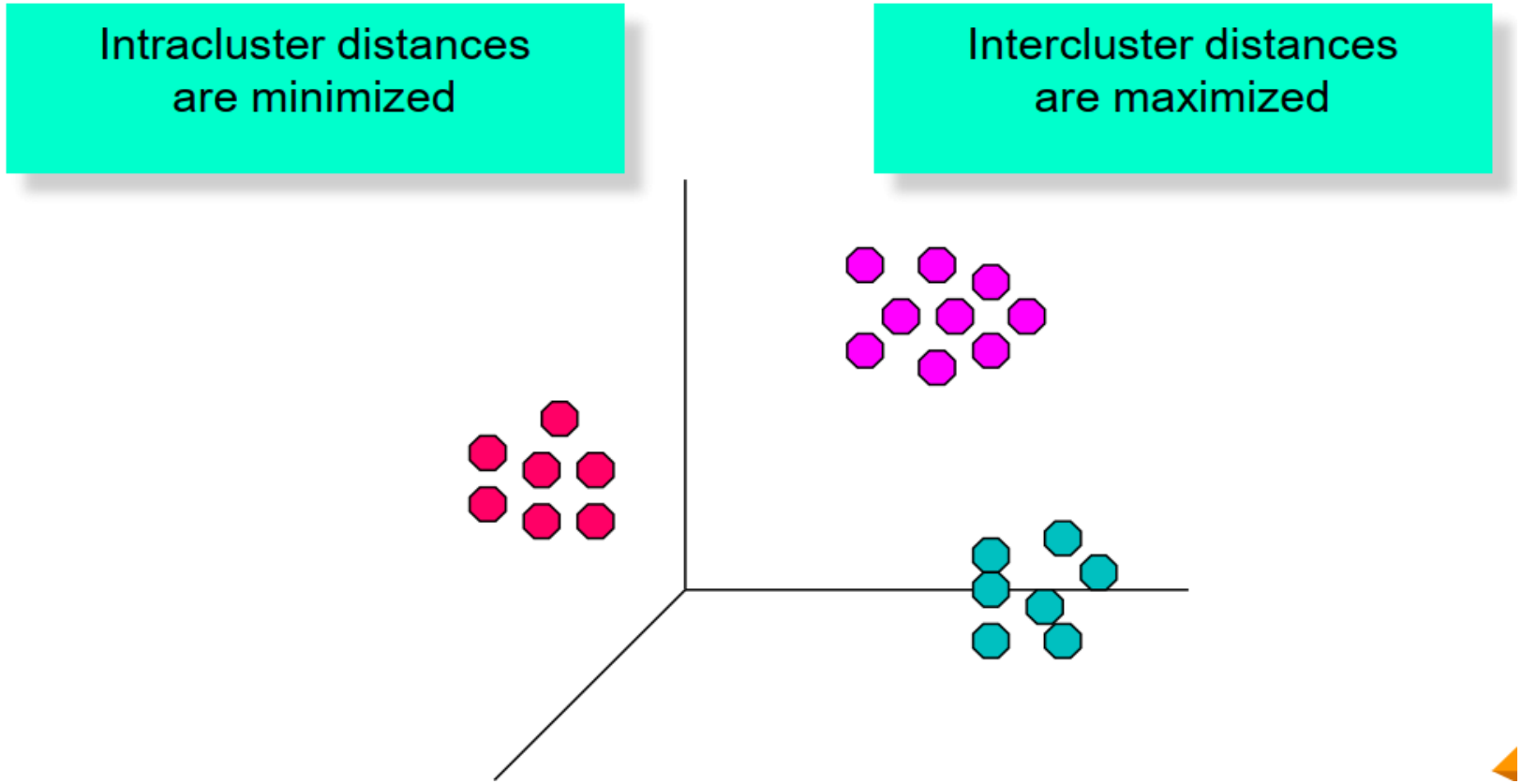
Clustering

- Given a set of data points, each having a set of attributes, and a similarity measure among them, find groups such that
  - data points in one group are more similar to one another.
  - data points in separate groups are less similar to one another.
- Similarity Measures
  - Euclidean distance if attributes are continuous
  - other task-specific similarity measures
- Goals/principle
  - Intra-cluster distances are minimized
  - inter-cluster distances are maximized
- Result
  - A descriptive grouping of data points



Clustering

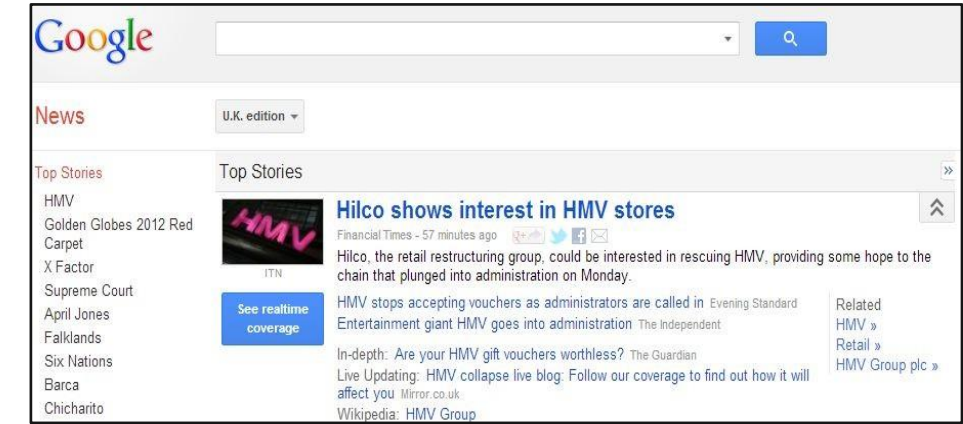
- Euclidean Distance Based Clustering in 3-D space.



- Principle: Maximizing intra-class similarity & minimizing interclass similarity

Clustering: Application 1

- **Application area: Document Clustering**
- Goal: Find groups of documents that are similar to each other based on terms appearing in them.
- Approach
  1. identify frequently occurring terms in each document
  2. form a similarity measure based on the frequencies of different terms. Use it to cluster
- Application Example: Grouping of articles in Google News



Illustrating Document Clustering

- **Clustering Points**: 3204 Articles of Los Angeles Times.
- **Similarity Measure:** How many words are common in these documents (after some word filtering).

Category	Total Articles	Correctly Placed
Financial	555	364
Foreign	341	260
National	273	36
Metro	943	746
Sports	738	573
Entertainment	354	278

Classification

- Given a collection of records (*training set*)
  - each record contains a set of *attributes*
  - one attribute is the *class attribute (label)* that should be predicted
- Find a *model (construct model)* for predicting the class attribute as a function of the values of other attributes.

Goal: Previously unseen records should be assigned a class from a given set of classes as accurately as possible.

- Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

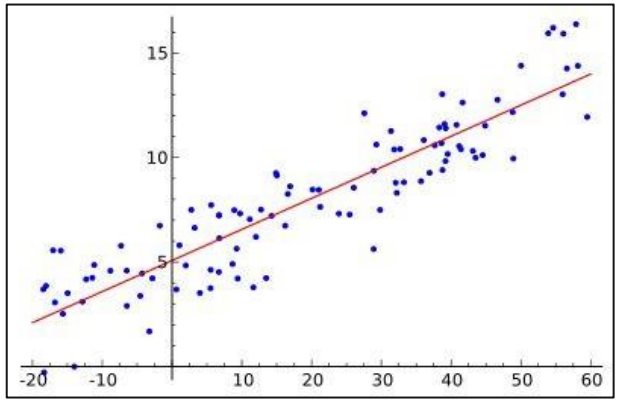


Classification : Application 1

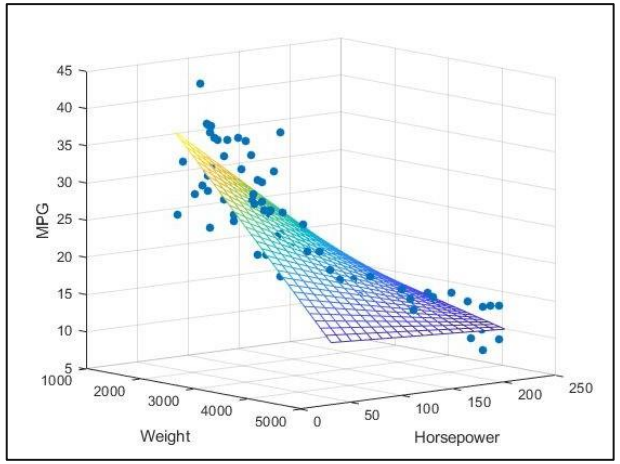
- **Direct Marketing**
- **Goal**: Reduce cost of mailing by targeting a set of consumers likely to buy a new cell-phone product.
- Approach:
  - Use the data for a similar product introduced before.
  - We know which customers decided to buy and which decided otherwise. This {buy, don't buy} decision forms **the class attribute**.
    - Collect various demographic, lifestyle, and company-interaction related information about all such customers.
    - age, profession, location, income, marriage status, visits, logins, etc.
    - Type of business, where they stay, how much they earn, etc.
  - Use this information as input attributes to learn a classification model
  - Apply model to decide which consumers to target

Regression

– Predict a value of a continuous variable based on the values of other variables, assuming a linear or nonlinear model of dependency.



– Examples:
  - Predicting the price of a house or car
  - Predicting wind velocities as a function of temperature, humidity, air pressure, etc.



– Difference to classification: The predicted attribute is continuous, while classification is used to predict nominal attributes (e.g. *yes/no*)

Association Rule Discovery

- Given a set of records each of which contain some number of items from a given collection.
- Discover frequent itemsets and produce association rules which will predict occurrence of an item based on occurrences of other items.
  - What items are frequently purchased together in a store?

TID	Items
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Frequent Itemsets
{Diaper, Milk, Beer}2
{Milk, Coke}3

Association Rules/Rules Discovered:
{Diaper, Milk} --> {Beer}
{Milk} --> {Coke}

Association Rule Discovery: Applications 1



- Application area: Supermarket shelf management.
  - Goal: To identify items that are bought together by sufficiently many customers
  - Approach: Process the point-of-sale data collected with barcode scanners to find dependencies among items
  - A classic rule and its implications:
    - if a customer buys diapers and milk, then he is likely to buy beer as well
    - so, don't be surprised if you find six-packs stacked next to diapers!
    - promote diapers to boost beer sales
    - if selling diapers is discontinued, this will affect beer sales as well
- Application area: Sales Promotion



Deviation /Anomaly Detection

- **Deviation detection:** identifying data points or patterns that significantly differ from the expected or normal behavior. These deviations, often referred to as **anomalies**, **outliers**, or **exceptions**, can indicate errors, fraud, system failures.



Applications:

- **Fraud detection:** Identifying unusual financial transactions.
- **Network Intrusion detection:** Detecting abnormal network traffic.
- **Customer behavior analysis:** Identifying unusual customer patterns.
- **System monitoring:** Detecting system failures or performance.

Typical network traffic at University level may reach over 100 million connections per day

“Questions ? “